

J3LE (Jelly) AI CyberLab

AI Infrastructure & Cybersecurity Research Lab

Project Vision

The J3LE AI CyberLab is a personal research and engineering environment focused on AI infrastructure, cybersecurity experimentation, and high-performance networking. The lab explores how modern enterprise networking architectures can support local AI inference, automation agents, and isolated cyber testing environments.

Core Research Areas

AI Infrastructure

Development of a local AI operator (J3LE / Jelly) designed to assist with system operations, automation workflows, and infrastructure management using local LLM models.

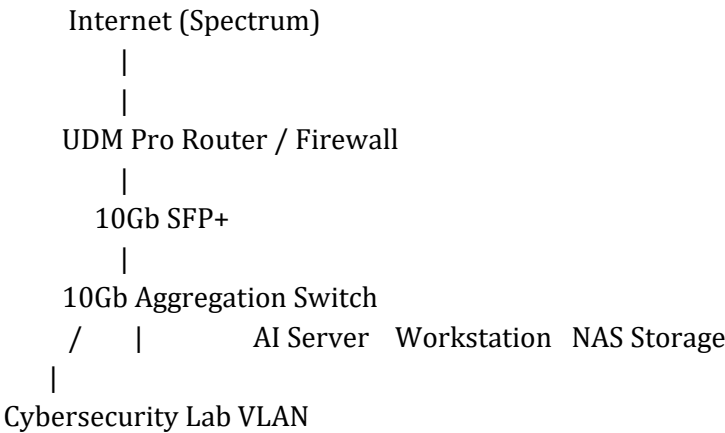
Cybersecurity Lab

A segmented cyber range used for security experimentation, vulnerability testing, and network attack simulations in an isolated environment.

High-Speed Networking

Implementation of a 10Gb internal backbone to support large AI model transfers, dataset storage, and distributed compute workloads.

Proposed Network Architecture



VLAN Segmentation Model

VLAN	Purpose
Home	Standard home devices
IoT	Smart devices and appliances
CyberLab	Security testing environment
AI	AI infrastructure and servers
Management	Network device administration

Publishing Strategy

To attract collaboration opportunities and sponsorship, the project will be published publicly across several platforms.

- GitHub repository documenting architecture, configurations, and diagrams.
- LinkedIn posts highlighting infrastructure experiments and lab builds.
- Ubiquiti Community posts demonstrating UniFi networking architectures.
- Technical blog posts documenting AI infrastructure experiments.

Recommended GitHub Repository Structure

```
J3LE-AI-CyberLab
|
├── README.md
├── docs
│   ├── network-architecture.md
│   ├── ai-infrastructure.md
│   └── cybersecurity-lab.md
├── diagrams
│   ├── network-topology.png
│   └── vlan-architecture.png
└── outreach
    └── vendor-collaboration.md
```

Vendor Collaboration Opportunities

The J3LE AI CyberLab seeks collaboration with companies focused on networking, AI infrastructure, storage, and cybersecurity technologies.

- Ubiquiti – networking infrastructure
- NVIDIA – GPU AI compute hardware
- Synology – storage platforms
- AMD – AI compute processors
- Cloudflare – security and edge networking

Future Infrastructure Roadmap

- Deploy 10Gb internal network backbone
- Add dedicated AI dataset storage
- Deploy GPU AI compute nodes
- Build Proxmox virtualization cluster
- Develop J3LE AI operator automation platform

Project Goals

The objective of the J3LE AI CyberLab is to create a practical environment for learning, experimentation, and publishing technical insights related to AI infrastructure and cybersecurity engineering.